Habit: A small annual herb

Root: Branched tap root system.

Stem: Aerial, erect, green and herbaceous

Leaf: Simple, alternate but opposite in the floral region, petiolate, exstipulate ovate, entire or slightly lobed, acute unicostate reticulate venation.

Inflorescence: Extra-axillary (due to fusion of floral axis) scorpioid cyme called rhiphidium

Flower: Ebracteate, pedicellate, white, bisexual, actinomorphic, heterochlamydeous, pentamerous, hypogynous white.

Calyx: Sepals 5, synsepalous, green, persistent and showing valvate aestivation.

Corolla: petals 5, sympetalous, white, showing valvate aestivation.

Androecium: Stamens 5, apostamenous, epipetalous, filaments short, anthers conniving and forming an envelope around the style dithecous, basifixed with apical pores.

Gynoecium: Bicarpellary, syncarpous, superior, bilocular, many ovules in each locule on axile placentation, septum oblique, highly swollen placenta, style long and hairy at the base, stigma bifid.

Fruit: Berry

Floral Formula: $\text{Ebr.,Ebrl.,}\oplus,\female,K_{(5)},\overset{\frown}{C_{(5)}},A_5,\underline{G}_{(2)}$

Economic importance

Economic importance of the family solanaceae				
S.No	**Economic importance**	**Binomial**	**Useful part**	**Uses**
1.	Food plant	*Solanum tuberosum* (potato)	Underground stem tubers	Used as vegetables and also used for the production of starch.
		Lycopersicon esculentum (tomato)	Ripened fruits	Used as delicious vegetable and eaten raw.
		Solanum melongena (brinjal)	Tender fruits	Cooked and eaten as vegetable.
		Capsicum annuum (bell peppers & chilli papers) *C. frutescens* (மிளகாய்)	Fruits	Used as vegetables and powdered chilli is the dried pulverized fruit which is used as spice to add pungency or piquancy and flavour to dishes .
		Physalis peruviana (cape gooseberry / சொடக்கு தக்காளி)	Fruit	Used as delicious fruit.